
11. Build Deployment

11.1 Introduction

Build Deployment is the process of preparing, packaging, and executing the automation framework along with the application under test in a controlled environment. It ensures that the latest version of the automation code is successfully built and executed without errors.

In this project, Build Deployment refers to the execution of the **Selenium WebDriver + TestNG automation framework** against the **OWASP Juice Shop Web Application** to validate security test cases such as SQL Injection, XSS, authentication testing, and input validation.

The build process is managed using **Maven**, which handles dependencies, compilation, and test execution.

11.2 Objective

The main objectives of Build Deployment are:

- To compile and build the automation framework successfully
 - To resolve all project dependencies using Maven
 - To execute test scripts using TestNG framework
 - To ensure integration between framework and OWASP Juice Shop application
 - To generate test execution reports automatically
 - To validate stability of automation scripts in execution environment
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11.3 Build Tool Used

The following build tool is used in this project:

- Maven (Apache Maven)

Maven is used because it provides:

- Dependency management
- Project structure standardization
- Automated build lifecycle
- Easy integration with TestNG and Selenium

11.4 Build Process Workflow

The build deployment process follows these steps:

Step 1: Code Compilation

- All Java source files are compiled
- Syntax and dependency errors are checked

Step 2: Dependency Resolution

- Maven downloads required libraries such as:
 - Selenium WebDriver
 - TestNG
 - WebDriver Manager (if used)

Step 3: Test Execution Trigger

- TestNG suite is triggered using `testng.xml`
- All test classes are executed in sequence

Step 4: Browser Execution

- Chrome browser is launched automatically
- OWASP Juice Shop application is opened
- Test scripts interact with UI elements

Step 5: Security Test Execution

The following security validations are performed during build execution:

- SQL Injection payload testing
 - Cross-Site Scripting (XSS) payload testing
 - Authentication validation
 - Input validation testing
 - Session handling verification
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Step 6: Report Generation

After execution, Maven and TestNG generate:

- `emailable-report.html`
 - `index.html`
 - `testng-results.xml`
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11.5 Build Execution Command

The build is executed using the following Maven command:

```
mvn clean test
```

Explanation:

- `clean` → removes previous build files
 - `test` → executes TestNG test cases
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11.6 Test Execution Flow During Build

1. TestNG reads `testng.xml`
 2. Test classes are loaded:
 - `Authentication_Test`
 - `SQLInjection_Test`
 - `XSS_Test`
 3. `BaseTest` initializes browser
 4. Test steps are executed sequentially
 5. Assertions validate expected results
 6. Results are logged and stored
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11.7 Output Generated After Build

After successful execution, the following outputs are generated:

Test Reports

- TestNG HTML Report (`index.html`)
- Emailable Report (`emailable-report.html`)

Logs

- Execution logs in console
- TestNG result XML file

Screenshots

- Captured using Screenshot Utility for failed test cases
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11.8 Build Validation

Build is considered successful when:

- All dependencies are resolved
 - No compilation errors exist
 - TestNG suite executes completely
 - All test cases run successfully
 - Reports are generated properly
 - Browser execution completes without crash
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11.9 Common Build Issues and Resolutions

Issue 1: Dependency Download Failure

- Cause: Internet or Maven repository issue
 - Fix: Update Maven or re-import project
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Issue 2: ChromeDriver Version Mismatch

- Cause: Browser and driver version mismatch
 - Fix: Update ChromeDriver version
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Issue 3: TestNG Execution Failure

- Cause: Incorrect testng.xml configuration
 - Fix: Verify class paths and suite structure
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11.10 Outcome

After build deployment:

- Automation framework executed successfully
 - Security test cases were triggered correctly
 - Reports and logs were generated
 - OWASP Juice Shop application was tested successfully
 - Framework stability was confirmed
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11.11 Conclusion

Build Deployment ensures that the automation framework is properly compiled, executed, and validated in a controlled environment. It plays a critical role in confirming that the security testing framework is stable, scalable, and ready for continuous execution.
